

Structural, morphological and photoluminescence properties of Ca_2SiO_4 : Er^{3+} phosphors synthesized from agro-food waste materials

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ABSTRACT

Trivalent erbium (Er^{3+})-embedded calcium silicate (Ca_2SiO_4) phosphors have been prepared by using the agro-food trash resources, such as egg shells and rice husk, via high temperature solid-state reaction technique and studied their structure, morphology and spectroscopic properties. X-ray diffraction patterns revealed that Ca_2SiO_4 phosphors are crystallized in monoclinic structure, while scanning electron microscopy (SEM) micrograph reflected that particles are agglomerated in different shapes and magnitudes with a non-uniform distribution. Luminescence spectra of Er^{3+} -doped Ca_2SiO_4 phosphors have been systematically investigated as a function of the Er^{3+} concentration under 488 and 980 nm laser excitations. Down shifting spectra exhibit characteristic emissions of Er^{3+} ions at around 524, 545 and 636 nm in the visible and at 1531 nm in the near-infrared regions. Non-exponential decay curves of the $^4\text{S}_{3/2}$ and $^4\text{I}_{13/2}$ emitting levels have been associated with decrease in lifetimes with increase in Er^{3+} ions concentration. Colour coordinates are shifted from the yellowish green to the green region with increase of Er^{3+} content. The phosphors also emit intense NIR-to-visible upconversion luminescence under 980 nm laser excitation. The above results indicate that the $\text{Ca}_2\text{SiO}_4:\text{Er}^{3+}$ phosphor could find potential applications in the WLEDs and optoelectronic devices.

1. Introduction

In the present era, several investigators have focused their attention to investigate new phosphors, which are the prime element of illumination and display accessories [1–3]. Pure white-light emitting diodes (WLEDs) are extensively investigated as future luminescence sources owing to their environment friendliness, long lifetime and high brightness. Commercial WLEDs are developed by combining a yellow emitting phosphor with blue emitting LED chip, although they are not useful for lighting applications due to their low colour rendering index [3,4]. The colour purity and luminescence efficiency of phosphors are pre-requisite parameters for design and development of plasma display panels. Solid state display technology is grounded on the utilization of phosphors in routine down converting, which can give visible and NIR radiation

under UV excitation [5]. Phosphors obtained by doping with certain trivalent lanthanide (Ln^{3+}) optically active ions can give visible emission under infrared laser excitation, a non-linear effect known as energy upconversion, are advantageous for display technologies or lighting. Infrared detectors, bio-labelling, compact solid state lasers, etc., are the important fields where upconverted luminescence of several Ln^{3+} ions doped phosphors are used [6–9].

Narrow-band emission under UV excitation and long lifetimes are significant features of the Ln^{3+} ions, and hence, they are being used as good activators in phosphors. Erbium (Er^{3+}) ion is one of the most interesting one among all the Ln^{3+} series because of its characteristic energy levels diagram, long lifetimes, which can be easily populated by exciting with diode lasers [10]. Hence, Er^{3+} ion is capable of achieving energy upconversion under 800, 980 and 1500 nm using low-cost laser

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diode excitations. Thus, Er^{3+} -embedded phosphors have gained much focus in the luminescent solar converters owing to their upconversion in near infrared luminescence.

Extreme luminescence efficiencies and high absorption under ultraviolet (UV) excitation are the important characteristics of Ln^{3+} -doped orthosilicate (Ca_2SiO_4) phosphors. In addition, these compounds carry dominating chemical and thermal stability and are used in plasma display panels, light-emitting devices, bio-molecules, sensors, bio-imaging, catalysts and as laser hosts [4,11–16].

Egg shells and rice husk are agro-food waste materials that are produced in huge amount every year which causes various environmental pollution as they are dumped into open fields. Conversion of the waste materials into wealth is attracting much interest of the scientific community all around the world. Rice husk is a potential material and one of the rich sources of silica [17–19]. It is extensively used in its either form of raw or ash materials, and a large portion of the husk is either dumped or burnt or as waste in out-door fields and modest quantity is utilized as energy for boilers, power production, bulking agents for composting of animal manure, etc. Additionally, egg shell, after taking inside matter, is a source for CaO [20]. Thus, agro-food waste material ashes are effectively utilized for numerous applications [17–20].

Few reports concerning Er^{3+} -doped phosphors have been found in the literature [21–39] and, to the best of our attention, there is no work on the synthesis of Er^{3+} -embedded Ca_2SiO_4 phosphors via solid state reaction technique by utilizing the agro-food resources, such as eggshell for CaO and rice husk for SiO_2 . In the present work, $\text{Ca}_2\text{SiO}_4:\text{Er}^{3+}$ phosphors have been synthesized from agro-food waste materials and studied their structural, morphological and photoluminescence characteristics.

2. Experimental details

2.1. Synthesis

The solid state reaction method was adopted for the syntheses of the Er^{3+} -doped Ca_2SiO_4 phosphors. The Er^{3+} -doping concentrations of the phosphors are: 0.01, 0.05, 0.1, 0.5 and 1.0 mol% and are labelled as $\text{CSEr}_{0.01}$, $\text{CSEr}_{0.05}$, $\text{CSEr}_{0.1}$, $\text{CSEr}_{0.5}$ and $\text{CSEr}_{1.0}$, respectively. The preparation process for extracting CaO (from egg shells) and SiO_2 (from rice husk) are described in our previous paper [40]. About 50 g of CaO has been yielded from 100 g of egg shell (raw material) with negligible amounts of impurities of carbon (C), magnesium (Mg) and silicon (Si), found from EDS analysis. Similarly, about 15 g of SiO_2 has been yielded from 100 g of rice husk (raw material) with negligible amounts of impurities of potassium (K), calcium (Ca), carbon (C), sodium (Na), magnesium (Mg) and phosphorous (P). The starting materials are taken in stoichiometric quantities. Calcium oxide (CaO) and silicon dioxide (SiO_2) were extracted from egg shells ashes and rice husk ashes, correspondingly, and Er_2O_3 was purchased from Sigma Aldrich (99.99%). All these chemicals were weighed in an electronic balance, transferred into an agate pestle and mortar and grinded in the medium of acetone. After grinding, the homogeneous fine powders were pre-sintered at 1200–1275 °C for 10 h with intermediate grindings and, eventually, sintered at 1200–1275 °C for 10 h. Final phosphors were developed by cooling naturally to room temperature.

2.2. Characterization

The structure of all the synthesized phosphors were analysed by using an X-ray diffractometer (RIGAKU, Miniflex-600) employing Cu- $\text{K}\alpha$ radiation (1.5406 Å). The surface morphologies of the present phosphors were examined by a field emission scanning electron microscopy (FE-SEM) and energy dispersive spectroscopy (EDS) was adopted to detect the elements present in the phosphors using a CARL ZEISS EVO MA15 SEM. Edinburgh (FLS-980) UV–visible–NIR spectrophotometer equipped with Xe arc lamp (450 W) was used as a pumping source to

record the excitation spectra. All the measurements were made at room temperature.

The visible (500–700 nm) and NIR luminescence spectra (1400–1700 nm) were studied by pumping the phosphors at 488 nm (Ar^+ laser, Beamlock Spectra Physics). Photoluminescence was studied through fiber coupled single grating (0.303 m) spectrograph (Andor Shamrock) and then detected visible emission with a cooled CCD detector and a near-infrared emission with an InGaAs detector. Upconverted and Stokes luminescence were recorded under a laser excitation of 980 nm using a Ti:sapphire laser pumping with $\text{Nd}^{3+}:\text{YAG}$ laser. The decay curves for the ($^2\text{H}_{11/2}$, $^4\text{S}_{3/2}$) and $^4\text{I}_{13/2}$ emitting levels of Er^{3+} ions were studied under laser excitations of 525 nm and 980 nm, respectively, using a 10 ns OPO and recorded with a digital storage oscilloscope coupled to a detection system.

3. Results and discussion

3.1. Structural studies

The XRD patterns for pure and 1.0 mol % Er^{3+} -embedded Ca_2SiO_4 phosphors are presented in Fig. 1 along with their (h k l) planes. XRD patterns which are well matched with ICDD card No. 01-083-0464, indicating that Er^{3+} -embedded Ca_2SiO_4 phosphors are in a monoclinic crystal structure and possess unit cell parameter values of $a = 3.955$ Å, $b = 4.006$ Å, $c = 5.80$ Å, $\beta = 103.5^\circ$, and a volume of ≈ 345 Å³. Also particle size is evaluated by applying Scherrer's Eq. [41], and is found to be ≈ 45 nm. It is also observed that there are no significant changes in the XRD patterns even with the increase of Er^{3+} content and, hence, the pattern is presented only for the $\text{CSEr}_{0.1}$ phosphor in Fig. 1. Structural parameters such as micro-strain (ϵ) = $\beta \cos(\theta/4)$ and dislocation density ($\delta = 1/D^2$), where ' β ' refers the full width at half maximum, ' θ ' corresponds to Bragg's angle and ' D ' indicates mean grain size of crystallites [42], are evaluated for $\text{CSEr}_{0.1}$ sample, and are found to be 48×10^{-5} and 6.38×10^{-2} nm, respectively.

3.2. Scanning electron microscopy and elemental analysis

The morphology of all synthesized Er^{3+} -doped Ca_2SiO_4 phosphors are recorded by SEM and its micrograph for 1.0 mol % Er^{3+} -doped phosphor is presented in Fig. 2. As can be seen, the particles are of different form and are distributed non-uniformly due to temperature gradient during their synthesis and calcinations process. Further, the elemental analysis is done by utilizing energy dispersive spectrum (EDS) for all agglomerated amalgamated phosphors and results are presented for the $\text{CSEr}_{1.0}$ phosphor in Fig. 3. From this, it is concluded that all the

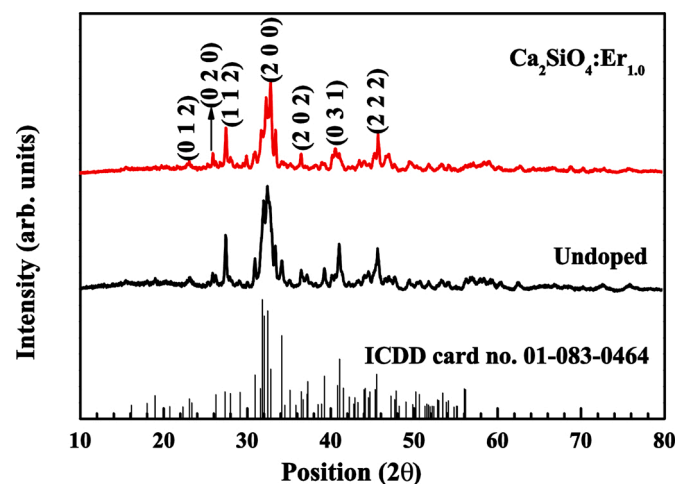


Fig. 1. X-Ray diffraction patterns and their (h k l) planes for undoped Ca_2SiO_4 and 1.0 mol% Er^{3+} -doped Ca_2SiO_4 phosphor.

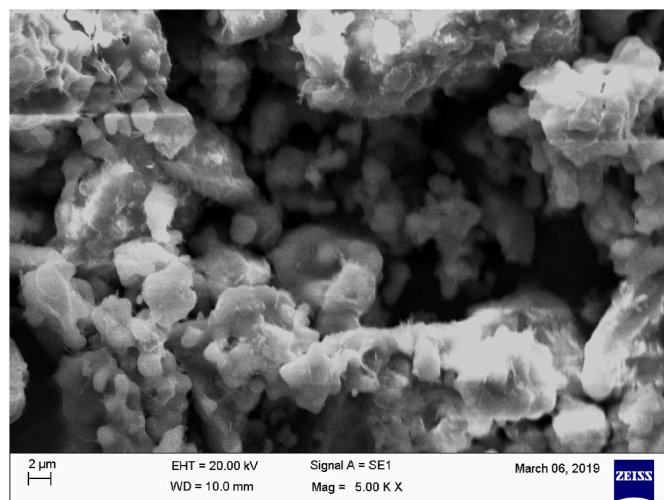


Fig. 2. Scanning electron micrograph of 1.0 mol % Er^{3+} -doped Ca_2SiO_4 phosphor.

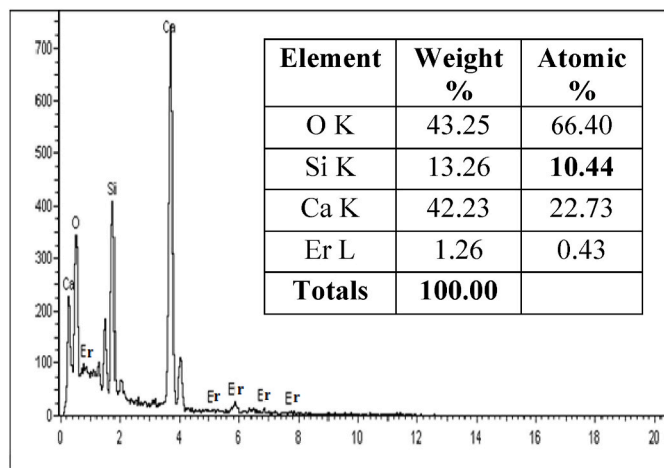


Fig. 3. Energy dispersive spectrum (EDS) of 1.0 mol % Er^{3+} -doped Ca_2SiO_4 phosphor.

elements exist in the phosphors that include oxygen (O), calcium (Ca), silica (Si) and erbium (Er), which are found in their individual ratios and also confirmed that no impurities are present in the final synthesized sample.

3.3. Photoluminescence studies

The photoluminescence excitation spectrum of $\text{Ca}_2\text{SiO}_4:\text{Er}^{3+}$ phosphor, measured by observing the emission at 545 nm is presented in Fig. 4. The spectrum consists of five peaks centered at around 366, 380, 407, 453 and 488 nm originating from the $^4\text{I}_{15/2}$ ground level to the $^4\text{G}_{9/2,11/2}$, $^2\text{G}_{9/2}$ and $^4\text{F}_{5/2,7/2}$ excited states of Er^{3+} ion [32]. The spectrum is devoid of any host absorption or charge transfer band in the shorter wavelength region. The transitions at 380, 453 and 488 nm exhibit highest intensity indicating that the phosphor could be suitably excited using the above wavelengths to get green component in tricolor-based WLEDs.

The emission spectra of Er^{3+} -doped Ca_2SiO_4 phosphors for different concentrations of Er^{3+} ions are recorded by exciting resonantly the $^4\text{I}_{15/2} \rightarrow ^4\text{F}_{7/2}$ level at 488 nm and are presented in Fig. 5. They consist of three emission peaks and among them an intense band is at 545 nm, related to the $^4\text{S}_{3/2} \rightarrow ^4\text{I}_{15/2}$ transition, and two feeble bands are at 523 nm and 639 nm, associated with the $^2\text{H}_{11/2} \rightarrow ^4\text{I}_{15/2}$ and $^4\text{F}_{9/2} \rightarrow ^4\text{I}_{15/2}$

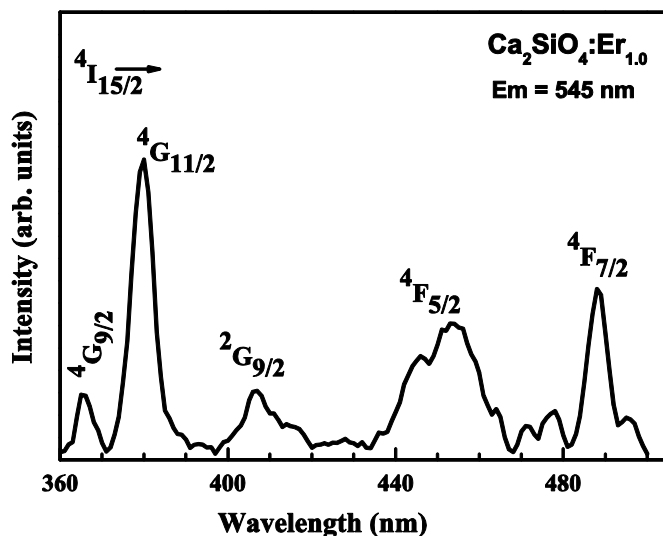


Fig. 4. Photoluminescence excitation spectrum of 0.1 mol % Er^{3+} -doped Ca_2SiO_4 phosphor, monitoring the emission at 545 nm.

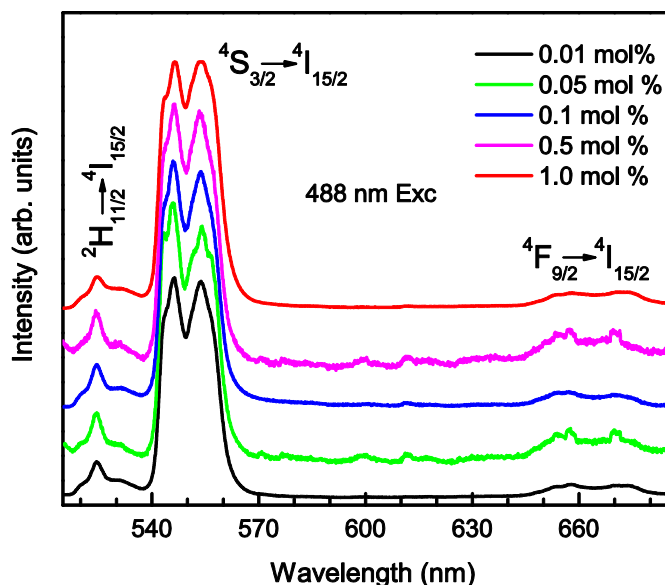


Fig. 5. Luminescence spectra of $\text{Ca}_2\text{SiO}_4:\text{Er}^{3+}$ phosphors in the visible region measured by exciting at 488 nm.

levels, respectively, although these bands further split into two or three peaks as a result of Stark splitting, due to the crystal-field interaction between the $4f^{d1}$ (Er^{3+}) electrons and the valence electrons of the oxygen ligands forming the nearest coordination sphere. The mechanisms of excitation and emission transitions are shown in the energy level diagram in Fig. 6. When exciting at 488 nm, the Er^{3+} ions are pumped to the $^4\text{F}_{7/2}$ state and then decay non-radiatively to the $^2\text{H}_{11/2}$, $^4\text{S}_{3/2}$, $^4\text{F}_{9/2}$ and $^4\text{I}_{13/2}$ levels giving rise to three radiative emissions in the visible region at 523, 545 and 639 nm and one in the NIR region (~ 1540 nm) as shown in Fig. 6.

It is also observed from Fig. 5 that the luminescence intensity is increasing with the increase in Er^{3+} concentration. The possible reason for this increase in intensity is an improvement in the phase purity of the phosphor with Er^{3+} concentration which leads to an enhancement in Er^{3+} radiative transition probabilities. Other possible explanation is charge compensation which results in higher local disorder and changes in the probabilities. This improves better distribution of Er^{3+} ions in the

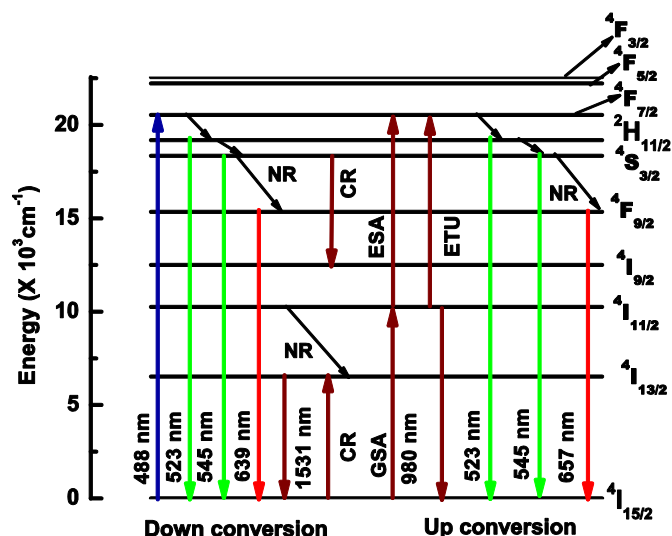


Fig. 6. Energy level diagram of Er³⁺ ions in Ca₂SiO₄ phosphors showing down and up conversion mechanisms (NR: Non-radiative decay; CR: Cross-relaxation; GSA: Ground state absorption; ESA: Excited state absorption; ETU: Energy transfer upconversion).

Ca₂SiO₄ phosphor and accounts for the enhanced luminescence with maximum intensity at 1.0 mol %. As can be seen from Fig. 5, there is a gradual increase in luminescence intensity from 0.01 to 1.0 mol % of Er³⁺, with no concentration quenching. This behaviour happens due to the fact that Er³⁺ (*r* = 96 p.m.) ions are replacing Ca²⁺ (*r* = 240 p.m.) in Ca₂SiO₄ structure [43] without decreasing luminescence intensity.

NIR luminescence spectra of the ⁴I_{13/2} → ⁴I_{15/2} (1.5 μm) level of different content of Er³⁺-embedded Ca₂SiO₄ phosphors are studied under 488 nm laser pumping and are depicted in Fig. 7. There is a splitting in this broad band, centred at around 1498, 1531, 1540 and 1555 nm and the maximum intensity is observed for the peak at 1531 nm. The intensity of the peak at 1540 nm increases gradually with increase in Er³⁺ ions concentration from 0.01 to 1.0 mol % as well as the full width at half-maximum (FWHM) of the emission band increases from 65 to 72 nm in this concentration range. Also, splitting of the bands increases from 0.01 to 0.1 mol % and beyond 0.1 mol % it decreases. The emission band at 1531 nm has extremely significant usage in optical communication systems [44].

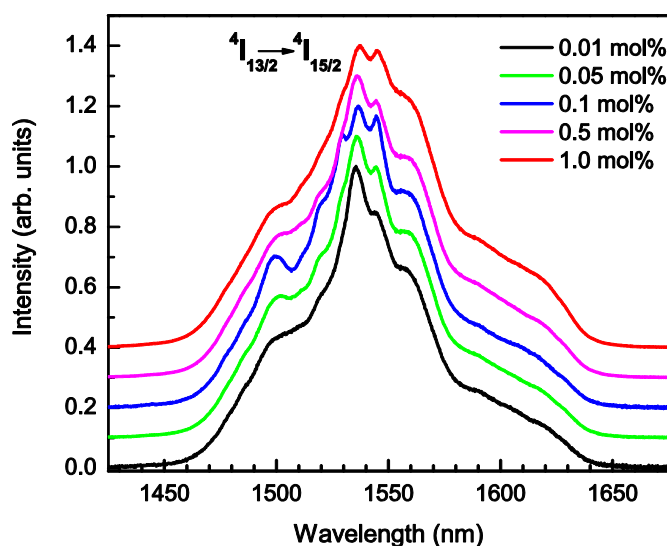


Fig. 7. Luminescence spectra of Ca₂SiO₄:Er³⁺ phosphors in the NIR region, measured by exciting at 488 nm.

Infrared-to-visible upconverted (UC) photoluminescence spectra of CSEr_{1.0} phosphors are studied under 980 nm laser pumping and are presented in Fig. 8. As seen from this figure, Er³⁺-doped Ca₂SiO₄ phosphors show intense green and weak red emissions. The UC spectra consists of three peaks and resemble the Stokes luminescence obtained under UV pumping.

The phenomena of UC can be described as follows: when pumped with 980 nm laser beam, the Er³⁺ in the ground state (⁴I_{15/2}) absorb photons and in turn excited to the ⁴I_{11/2} state through ground state absorption (GSA). The Er³⁺ ions in the ⁴I_{11/2} level further absorb photons from the excitation source and are pumped to the ⁴F_{7/2} level through a process called excited state absorption (ESA) and is shown in Fig. 6. In another process, the Er³⁺, present in ⁴I_{11/2} state goes to the ground state, transferring the emitted energy non-radiatively to other Er³⁺ ions present in the ⁴I_{11/2} level, pumping them to the ⁴F_{7/2} level, called energy transfer upconversion (ETU: ⁴I_{11/2} + ⁴I_{11/2} → ⁴F_{7/2} + ⁴I_{15/2}) as presented in Fig. 6. The Er³⁺ in the ⁴F_{7/2} state relax non-radiatively and populate the (²H_{11/2}, ⁴S_{3/2}) emitting states. From there, radiative transition to the ground state, ⁴I_{15/2} takes place giving rise to green fluorescence, ²H_{11/2} → ⁴I_{15/2} (523 nm) and ⁴S_{3/2} → ⁴I_{15/2} (545 nm). As the ⁴S_{3/2} state is close to the ⁴F_{9/2} state, some of the Er³⁺ relax non-radiatively by multi-photon relaxation to the ⁴F_{9/2} state. The Er³⁺ ions in the ⁴F_{9/2} state then reach the ground level (⁴F_{9/2} → ⁴I_{15/2}) emitting photons at the wavelength of 657 nm.

In order to evaluate the number of NIR laser photons responsible in the upconversion emission phenomena, a log-log plot of the luminescence intensity versus laser power has been drawn (shown as inset of Fig. 8). The slopes are determined to be 1.4 and 2.0 for green and red emissions, respectively, indicating that the process of UC is a two photon process [45].

3.4. CIE colour co-ordinates

The Commission Internationale de l'Eclairage (CIE) 1931 chromaticity diagram for Ca₂SiO₄:Er³⁺ phosphors is shown in Fig. 9. The colour coordinates for various concentrations of Er³⁺ ions-doped phosphors of the present study are evaluated and are shown in Table 1. As can be seen in Fig. 9, emission colour changes from yellowish-green to green with the increase in Er³⁺ concentration. From these co-ordinates, correlated color temperature (CCT) values are computed using the McCamy's approximate method [46] given by,

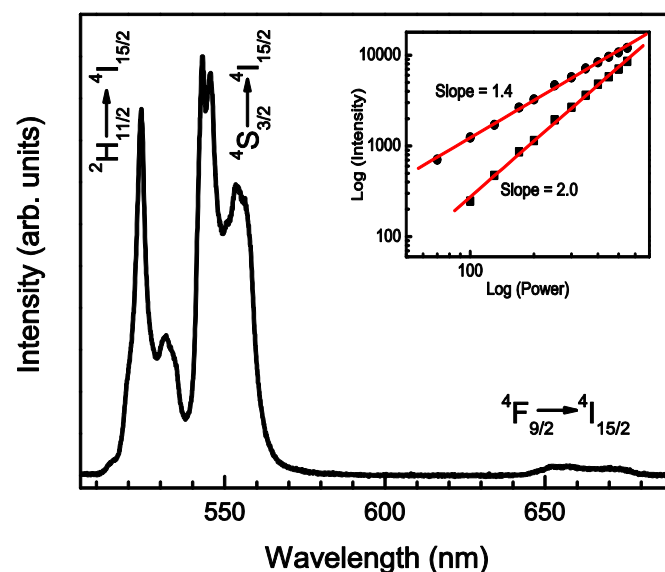


Fig. 8. Upconversion spectra of 1 mol % Er³⁺-doped Ca₂SiO₄ phosphors under 980 nm excitation. Inset shows the pump power dependence of upconversion emission intensity.

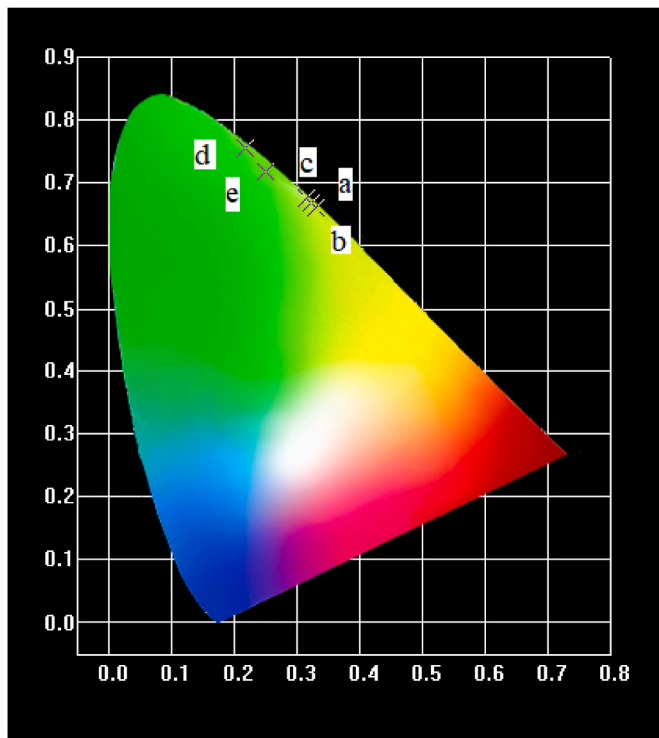


Fig. 9. CIE chromaticity diagram for $\text{Ca}_2\text{SiO}_4:\text{Er}^{3+}$ phosphors.

Table 1

CIE chromaticity coordinates (x,y), correlated color temperature (CCT, K) color purity (CP,%) lifetimes (τ , μs) and energy transfer parameters (Q) for different concentrations (C, mol %) of Er^{3+} -doped Ca_2SiO_4 phosphors under 488 nm excitation.

S. No.	C	(x, y)	CCT	CP	τ	Q
1	0.01	(0.322, 0.668)	5650	98	4	0.620
2	0.05	(0.331, 0.661)	5527	96	–	–
3	0.1	(0.314, 0.676)	5768	99	3.5	0.776
4	0.5	(0.218, 0.754)	7035	124	–	–
5	1.0	(0.251, 0.718)	6642	112	2.7	0.916

$$\text{CCT} = -449n^3 + 3525n^2 - 6823n + 5520.33 \quad (1)$$

Here 'n' is equal to $(x-x_e)/(y-y_e)$ and ($x_e = 0.332$ and $y_e = 0.186$) which are the epicentre of the convergence. The CCT values for the synthesized samples are listed in Table 1 and are found to be more than 5000 K indicating that the present phosphors are suitable for making cool W-LEDs.

The CIE coordinates are also used to obtain the colour purity (CP) of a phosphor. The CP is an important element for calculating the application of phosphors and its value can be evaluated employing the following Eq. [20],

$$\text{Colour purity} = \frac{\sqrt{(x-x_i)^2 + (y-y_i)^2}}{\sqrt{(x_d-x_i)^2 + (y_d-y_i)^2}} \times 100\% \quad (2)$$

Here (x, y) represents the CIE co-ordinates equivalent to the standard sample point and (x_i, y_i) stands for the illuminant point of CIE co-ordinates. The co-ordinates (x_d, y_d) are related to the dominant wavelength. The evaluated CP values of the present phosphors are shown in Table 1.

The CP values of the present phosphors ranges from 96 to 124 which are higher than those of the many systems that are compared in Table 2.

Table 2

Comparison of CIE chromaticity coordinates (x, y), correlated colour temperature (CCT, K), colour purity (CP, %) and lifetimes (τ , μs) for Er^{3+} -doped phosphors.

S. No.	Host (Er^{3+} mol %)	(x, y)	CCT	CP	τ
1.	$\text{CSr}_{1.0}$ (Present work)	(0.251, 0.718)	6642	112	2.7
2.	$\text{K}(\text{PO}_3)_4$ [21]	(0.28, 0.7)	6247	107	–
3.	$\text{Bi}_4\text{Ti}_3\text{O}_{12}$ (0.3) [22]	(0.31, 0.68)	5831	101	–
4.	BaGd_2O_4 : (0.04) [23]	(0.35, 0.64)	5255	91	–
5.	Y_2O_3 : (2.0) [24]	(0.30, 0.64)	6018	90	–
6.	$\text{CaLa}_2\text{ZnO}_5$: (1) [25]	(0.31, 0.66)	5845	95	20.5
7.	CaB_2O_4 : (0.08) [26]	(0.44, 0.53)	3656	91	6.9
8.	CPB: (0.6) [27]	(0.29, 0.59)	6268	99	8.1
9.	BaTiO_3 : (0.5) [28]	(0.43, 0.56)	3966	75	104
10.	Ba_2CaWO_6 : (0.20) [29]	(0.36, 0.58)	5073	86	6
11.	Gd_2O_3 : (0.5) [30]	(0.32, 0.39)	5934	21	–
12.	$\text{K}_{0.5}\text{Na}_{0.5}\text{NbO}_3$: (1.0) [31]	(0.27, 0.72)	6361	113	122
13.	$\text{BaGd}_{1.95}(\text{MoO}_4)_4$: (0.07) [32]	(0.23, 0.61)	7372	85	–
14.	YAlO_3 : (2.0) [33]	(0.31, 0.53)	5971	59	–
15.	$\text{HNH}_4\text{P}_3\text{O}_{10}$ [34]:	(0.54, 0.45)	2113	74	15.4
16.	Sr_2CeO_4 : (0.5) [35]	(0.30, 0.68)	5977	101	–
17.	CaY_2O_4 : (1.0) [36]	(0.36, 0.56)	5029	69	–

Higher CP values indicate that the phosphors are suitable for generation of pure green colour [44]. European Broadcasting Union (EBU) says that CIE 1931 coordinates of (0.29, 0.60) corresponds to standard green illumination [44]. According to National TV system committee (NTSC) primary green colour corresponds to CIE 1931 coordinates of (0.21, 0.71) [44]. As the colour coordinates of the present phosphors are nearer to the standard ones, they are very much useful for green light generation and in white LEDs as a green component [47].

The $\text{Ca}_2\text{SiO}_4:\text{Er}^{3+}$ phosphors synthesized from farming scrap such as egg shells and rice husk produce green light emission [28,29]. The CCT and CP values are significant, compared to those of reported Er^{3+} -doped phosphors and are shown in Table 2 [21–36]. Hence, $\text{Ca}_2\text{SiO}_4:\text{Er}^{3+}$ phosphors can act as cost effective potential green light emitting phosphors.

3.5. Decay of the luminescence

All the visible emission decay profiles for the $^4\text{S}_{3/2}$ state of Er^{3+} : Ca_2SiO_4 phosphors studied under 525 nm pumping, by observing 545 nm emission, are shown in Fig. 10. The decay curves show increase in non-exponential character with increase in Er^{3+} content. Hence,

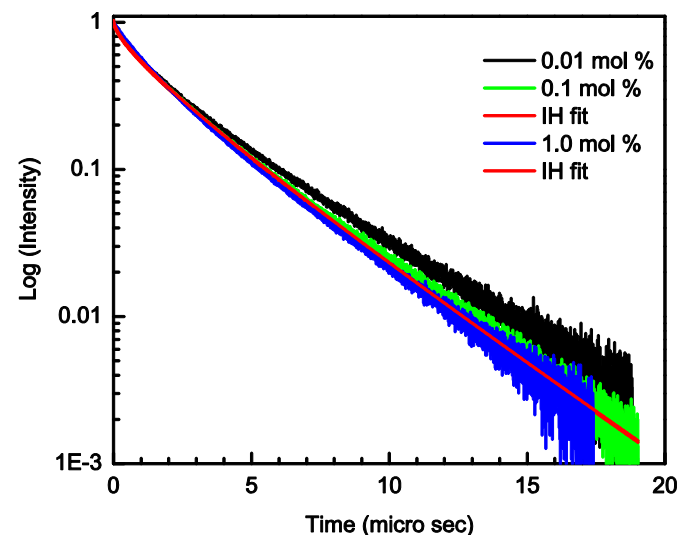


Fig. 10. Decay curves of the $^4\text{S}_{3/2}$ level for different concentrations of Er^{3+} ions in $\text{Ca}_2\text{SiO}_4:\text{Er}^{3+}$ phosphors along with IH model fit for 0.1 and 1.0 mol % curves.

effective lifetimes (τ_{eff}) are calculated by using the following Eq.,

$$\tau_{\text{eff}} = \frac{\int tI(t)dt}{\int I(t)dt} \quad (3)$$

Lifetimes of the $^4\text{S}_{3/2}$ state of Er^{3+} ions in Ca_2SiO_4 phosphors are found to decrease from 4 to 2.7 μs when Er^{3+} content increases from 0.01 to 1.0 mol % and are shown in Table 1. The shortening of lifetimes of Er^{3+} ions is due to existence of non-radiative energy transfer between Er^{3+} ions through cross-relaxation process ($^4\text{S}_{3/2}, ^4\text{I}_{15/2} \rightarrow ^4\text{I}_{9/2}, ^4\text{I}_{13/2}$). Similar behaviour of lifetimes of the $^4\text{S}_{3/2}$ level with increase in Er^{3+} concentration was also observed in silica based phosphors [48]. Lifetime values of the $^4\text{S}_{3/2}$ state of Er^{3+} ions in various phosphors [25–29,31,34] are listed in Table 2.

The non-exponential character of decay profiles of Er^{3+} -doped Ca_2SiO_4 phosphors, can be described by Inokuti-Hirayama (IH) model [49]. As such, the decay intensity $I(t)$ is expressed as,

$$I(t) = I_0 \exp \left\{ -\frac{t}{\tau_0} - Q \left(\frac{t}{\tau_0} \right)^{3/S} \right\} \quad (4)$$

Here 't' indicates the decay time after excitation and τ_0 refers the intrinsic decay time of the donor in the absence of acceptor. In the above Eq., 'S' value gives the nature of interaction mechanism between the Er^{3+} ions. So, 'S' may closely fit to 6 (dipole-dipole), 8 (dipole-quadrupole) or 10 (quadrupole-quadrupole) depending on the interaction mechanism. The decay profiles of Er^{3+} -doped Ca_2SiO_4 phosphors are best fitted to $S = 6$ of Eq. (4) implying that the interaction between Er^{3+} ions is of dipole-dipole nature. The IH model fits are shown in Fig. 10 for the decay curves of $\text{CSEr}_{0.1}$ and $\text{CSEr}_{1.0}$ phosphors.

The 'Q' value determines the energy transfer between Er^{3+} ions and its value, obtained for all Er^{3+} ions contents of the Ca_2SiO_4 phosphors, are listed in Table 1. From these values it is clear that 'Q' value increases with increase in Er^{3+} ions content. Hence, energy transfer between the Er^{3+} is responsible for decreasing lifetime values with increase in concentration.

Lifetimes of the $^4\text{I}_{13/2}$ state are determined for various contents (0.01, 0.1 and 1.0 mol %) of Er^{3+} in Ca_2SiO_4 phosphors, exciting at 488 nm and are shown in Fig. 11. All the three decay curves are non-exponential in nature and their lifetimes are determined using Eq. (3) and are found to 370, 52 and 25 μs for 0.01, 0.1 and 1.0 mol % Er^{3+} -doped phosphors, respectively. The decrease in lifetimes of the $^4\text{I}_{13/2}$ state with increase in Er^{3+} content is because of concentration quenching arising from non-radiative energy transfer processes between the nearby Er^{3+} ions and then to traps.

4. Conclusions

The Er^{3+} -embedded calcium silicate phosphors have been prepared using agro-food waste products through solid state reaction technique and studied their structure, morphology and luminescence characteristics. The synthesized phosphors have monoclinic crystal structure with agglomerated morphology of different shapes with irregular sizes in nature. Excitation spectrum of the synthesized phosphors, monitoring the emission at 545 nm, exhibits most intense bands at 380, 453 and 488 nm. The present phosphor shows luminescence in the green ($^2\text{H}_{11/2} \rightarrow ^4\text{I}_{15/2}$ and $^4\text{S}_{3/2} \rightarrow ^4\text{I}_{15/2}$) and red ($^4\text{F}_{9/2} \rightarrow ^4\text{I}_{15/2}$) spectral regions under both 488 nm (down conversion) and 980 nm (upconversion) excitations. The CIE colour coordinates for the present phosphors shift from yellowish green to green colour region under 488 nm excitation and the respective CP quantities have been evaluated. The higher CP values indicate that the present phosphors are suitable for generation of pure green colour. The decay curves of the emitting levels exhibit non-exponential nature for all concentrations due to dipole-dipole interaction between the Er^{3+} ions. The results show that the present phosphors could be suitable for WLEDs and optoelectronic devices.

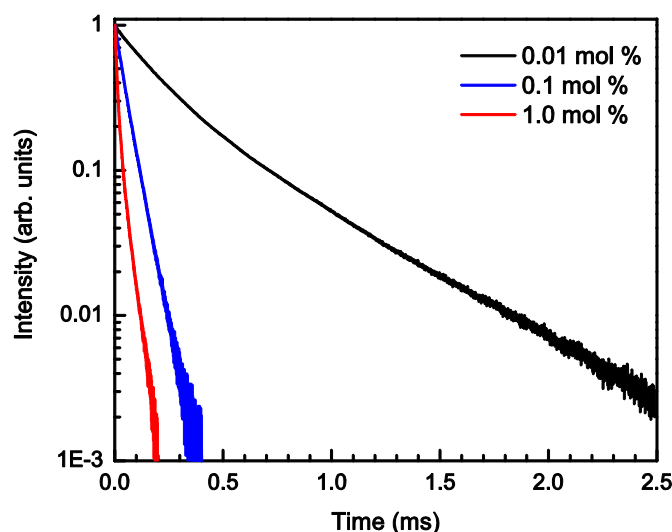


Fig. 11. Decay curves of the $^4\text{I}_{13/2}$ level for different concentration of Er^{3+} ions in $\text{Ca}_2\text{SiO}_4:\text{Er}^{3+}$ phosphors.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

- [1] R.G. Ye, H.P. Ma, C. Zhang, Y.Q. Gao, Y.J. Hua, D.G. Deng, P. Liu, S.Q. Xu, Luminescence properties and energy transfer mechanism of $\text{Ce}^{3+}/\text{Mn}^{2+}$ co-doped transparent glass-ceramics containing $\beta\text{-Zn}_2\text{SiO}_4$ nano-crystals for white light emission, *J. Alloys Compd.* 566 (2013) 73–77.
- [2] X.F. Li, J.D. Budai, F. Liu, J.Y. Howe, J.H. Zhang, X.J. Wang, Z.J. Gu, C.J. Sun, R. S. Meltzer, Z.W. Pan, New yellow $\text{Ba}_{0.93}\text{Eu}_{0.07}\text{Al}_2\text{O}_4$ phosphor for warm-white light-emitting diodes through single-emitting-center conversion, *Light Sci. Appl.* 2 (2013) 50–58.
- [3] B.V. Ratnam, M. Jayasimhadri, G.B. Kumar, K. Jang, S.S. Kim, Y.I. Lee, J.M. Lim, D. S. Shin, T.K. Song, Synthesis and luminescent features of $\text{NaCaPO}_4:\text{Tb}^{3+}$ green phosphor for near UV-based LEDs, *J. Alloys Compd.* 564 (2013) 100–104.
- [4] E.F. Schubert, *Light Emitting Diodes*, second ed., Cambridge University Press, New York, 2006.
- [5] V. Singh, V.K. Rai, M. Haase, Intense green and red upconversion emission of Er^{3+} , Yb^{3+} co-doped CaZrO_3 obtained by a solution combustion reaction, *J. Appl. Phys.* 112 (2012), 063105.
- [6] P. Sudhakar, A. Siva Sessa Reddy, Ya Zhyachevskyy, A. Suchocki, M.G. Brike, V. Ravi Kumar, M. Piasecki, N. Veeraiah, Luminescence characteristics of Er^{3+} ions in $\text{ZnO-Ta}_2\text{O}_5/\text{Nb}_2\text{O}_5/\text{ZrO}_2\text{-B}_2\text{O}_3$ glass system- A case study of energy transfer from ZnO to Er^{3+} ions, *Opt. Mater.* 86 (2018) 87–94.
- [7] J. Janek, R. Lisiecki, W. Ryba-Romanowski, J. Pisarska, W.A. Pisarski, Up-conversion luminescence of Er^{3+} ions in lead-free germinate glasses under 800 nm and 980 nm cw diode laser excitation, *Opt. Mater.* 74 (2017) 105–108.

- [8] E.W. Barrera, Q. Madueño, F.J. Novegil, A. Speghini, M. Bettinelli, Luminescence upconversion of Er:Yb:KY(WO₄)₂ green phosphor with high color purity, *Opt. Mater.* 84 (2018) 354–359.
- [9] P.P. Mokoena, H.C. Swart, O.M. Ntwaeaborwa, Upconversion luminescence of Er³⁺/Yb³⁺ doped Sr₅(PO₄)₃OH phosphor powders, *Physica B* 535 (2018) 57–62.
- [10] D.C. Zhou, Z.G. Song, G.W. Chi, J.B. Qiu, NIR broadband luminescence and energy transfer in Er³⁺-Tm³⁺-co-doped tellurite glasses, *J. Alloys Compd.* 481 (2009) 881–884.
- [11] O. Guillot-Noel, B. Vianna, B. Bellamy, Spectroscopic evidence of inhomogeneous distribution of Nd³⁺ in YVO₄, YPO₄ and YAsO₄ crystals, *Opt. Mater.* 13 (2000) 427–437.
- [12] R.S. Meltzer, S.P. Feofilov, Spectral hole burning in the 4f-5d transition of Ce³⁺ in LuPO₄ and YPO₄, *J. Lumin.* 102 (2003) 151–155.
- [13] I. Hemmila, Luminescent lanthanide chelates away to more sensitive diagnostic methods, *J. Alloys Compd.* 225 (1995) 480–485.
- [14] W. Di, X. Wang, B. Chen, H. Lai, X. Zhano, Preparation, characterization and VUV luminescence property of YPO₄:Tb phosphor for a PDP, *Opt. Mater.* 27 (2005) 1386–1390.
- [15] P. Yang, Z. Quan, C. Li, Z. Hou, W. Wang, J. Lin, Solvothermal synthesis and luminescent properties of monodisperse LaPO₄:Ln (Ln=Eu³⁺, Ce³⁺, Tb³⁺) particles, *J. Solid State Chem.* 182 (2009) 1045–1054.
- [16] R. Balakrishnaiah, D.W. Kim, S.S. Yi, S.H. Kim, K. Jang, H.S. Lee, B.K. Moon, J. H. Jeong, NIR to VIS frequency upconversion luminescence properties of Er³⁺-doped YPO₄ phosphors, *Thin Solid Films* 518 (2010) 6145–6148.
- [17] J. Kaewkhao, P. Limsuwan, Utilization of rice husk fly ash in the color glass production, *Procedia Eng.* 32 (2012) 670–675.
- [18] T. Lee, R. Othman, F.Y. Yeoh, Development of photoluminescent glass derived from rice husk, *Biomass. Bioenergy* 59 (2013) 380–392.
- [19] C.S. Prasad, K.N. Maiti, R. Venugopal, Effect of substitution of quartz by rice husk ash and silica fume on the properties of whiteware compositions, *Ceram. Int.* 29 (2003) 907–914.
- [20] N. Viriya-empikul, P. Krasae, B. Puttasawat, B. Yoosuk, N. Chollacoop, K. Faungnawakij, Waste shells of mollusk and egg as biodiesel production catalysts, *Bioresour. Technol.* 101 (2010) 3765–3767.
- [21] J.C. Mzali, K.H. Naifer, M. Férid, Investigations of vibrational, structural and optical properties of erbium polyphosphate micro-powders, *Optik* 127 (2016) 6340–6350.
- [22] B. Renuka, O.P. Thakur, V.K. Rai, S.K. Sharma, K. Sreenivas, Dielectric, ferroelectric and photoluminescence properties of Er³⁺-doped Bi₄Ti₃O₁₂ ferroelectric ceramics, *Ceram. Int.* 41 (2015) 6055–6066.
- [23] P. Du, J.S. Yu, Doping concentration-independent optical thermometric properties in Stark sublevels-based Er³⁺-activated BaGd₂O₄ luminescent thermometers, *J. Lumin.* 203 (2018) 172–178.
- [24] V. Dubey, R. Tiwari, R.K. Tamrakar, G.S. Rathore, C. Sharma, N. Tiwari, Infrared spectroscopy and upconversion luminescence behaviour of erbium doped yttrium (III) oxide phosphor, *Infrared Phys. Technol.* 67 (2014) 537–541.
- [25] Sk Khaja Hussain, G. Seeta Rama Raju, J.S. Yun, Synthesis and luminescent properties of CaLa₂ZnO₅:Ln (Ln:Tm³⁺ or Er³⁺) phosphors, *Ceram. Int.* 41 (2015) 13264–13270.
- [26] P. Remya Mohan, Subash Gopi, Viji Vidyadharan, Anns George, Cyriac Joseph, N. V. Unnikrishnan, P.R. Biju, Synthesis and luminescence characteristics of CaB₂O₄:Er³⁺, Li⁺ Phosphor, *J. Lumin.* 187 (2017) 113–120.
- [27] V. Reddy Prasad, B. Hariitha, S. Damodaraiah, Y.C. Ratnakaram, Influence of Nd³⁺ and Er³⁺ concentration on NIR luminescence properties in calcium borophosphate (CBP) phosphors, *Infrared Phys. Technol.* 94 (2018) 184–190.
- [28] D.K. Singh, J. Manam, Efficient dual emission mode of green emitting perovskite BaTiO₃:Er³⁺ phosphors for display and temperature sensing applications, *Ceram. Int.* 44 (2018) 10912–10920.
- [29] E. Sreeja, P. Remya Mohan, Subash Gopi, Cyriac Joseph, N.V. Unnikrishnan, P. R. Biju, Structural and photoluminescence properties of UV-excited Er³⁺ doped Ba₂CaWO₆ yellowish-green phosphors, *Phys. B Condens. Matter* 555 (2019) 74–80.
- [30] R.K. Tamrakar, D.P. Bisen, K. Upadhyay, I.P. Sahu, Upconversion and colour tunability of Gd₂O₃:Er³⁺ phosphor prepared by combustion synthesis method, *J. Alloys Compd.* 655 (2016) 423–432.
- [31] X. Wu, C.M. Lau, K.W. Kwok, Effect of phase transition on photoluminescence of Er-doped KNN ceramics, *J. Lumin.* 155 (2014) 343–350.
- [32] Y. Deng, S. Yi, Y. Wang, J. Xian, Synthesis and photoluminescence characteristics of Ln³⁺ (Ln = Sm, Er and Dy)-doped BaGd₂(MoO₄)₄ phosphors, *Opt. Mater.* 36 (2014) 1378–1383.
- [33] H.N. Baig, J.K. Saluja, S.J. Dhoble, Synthesis and characterization of erbium-doped YAlO₃ phosphor, *Luminescence* 31 (2016) 401–406.
- [34] J.Ch Mzali, S. Elgharbi, K.H. Naifer, M. Ferid, Synthesis, vibrational characteristic and luminescence properties of Er³⁺ in HNH₄ErP₃O₁₀, *Opt. Mater.* 73 (2017) 188–196.
- [35] K.V.R. Murthy, B. Nageswara Rao, B.N. Rajasekhar, B. Walter Ratna Kumar, K. Suresh, B. Subba Rao, Effect of Flux on the formation of Sr₂CeO₄ phosphor doped with Er, *Phys. Procedia* 29 (2012) 65–69.
- [36] V. Dubey, R. Tiwari, R.K. Tamrakar, Ch Markande, G.S. Rathore, M.K. Pradhan, Thermoluminescence and photoluminescence study of erbium doped CaY₂O₄ phosphor, *Ind. J. Mater. Sci.* 2015 (2015), 745052.
- [37] Y. Zhao, G. Bai, Y. Huang, Y. Liu, D. Peng, L. Chen, S. Xu, Stimuli responsive lanthanide ions doped layered piezophotonic microcrystals for optical multifunctional sensing applications, *Nano Energy* 87 (2021), 106177.
- [38] K. He, L. Zhang, Y. Liu, B. Xu, L. Chen, G. Bai, Lanthanide ions doped nonhygroscopic La₂Mo₃O₁₂ microcrystals based on multimode luminescence for optical thermometry, *J. Alloys Compd.* 890 (2022), 161918.
- [39] Y. Luo, L. Zhang, Y. Liu, E. Heydari, L. Chen, G. Bai, Designing dual-mode luminescence in Er³⁺-doped Y₂WO₆ microparticles for anticounterfeiting and temperature measurement, *J. Am. Ceram. Soc.* 105 (2022) 1375–1385.
- [40] L. Lakshmi Devi, Ch Basavapoornima, V. Venkatramu, P. Babu, C.K. Jayasankar, Synthesis of Ca₂SiO₄:Dy³⁺ phosphors from agricultural waste for solid state lighting applications, *Ceram. Int.* 43 (2017) 16622–16627.
- [41] A.L. Patterson, The Scherrer formula for X-ray particle size determination, *Phys. Rev.* 56 (1939) 978–982.
- [42] D.V. Sunitha, H. Nagabhushana, S.C. Sharma, B.M. Nagabhushana, R.P. S. Chakradhar, Luminescent characteristics of Eu³⁺ doped di-calcium silicate nano-powders for white LEDs, *J. Alloys Compd.* 575 (2013) 434–443.
- [43] L. Lakshmi Devi, Ch Basavapoornima, V. Venkatramu, C.K. Jayasankar, J. Kaewkhao, W. Lertlop, Structural and luminescence properties of Sm³⁺-doped Ca₂SiO₄ phosphors from agricultural waste, *Mater. Today Proc.* 5 (2018) 15081–15085.
- [44] N.S. Prabhu, A.N. Meza-Rocha, O. Soriano-Romero, U. Caldino, E.F. Huerta, C. Falcony, M.I. Sayyed, H. Al-Ghamdi, A.H. Almuqrin, S.D. Kamath, Spectroscopic study of Er³⁺-doped borate glass system for green emission device, NIR laser and optical amplifier applications, *J. Lumin.* 238 (2021), 118216.
- [45] J. Chen, W. Zhang, S. Cui, X. Peng, F. Hu, R. Wei, H. Guo, D. Huang, Up-conversion luminescence properties and temperature sensing performances of Ba₅Y₈Zn₄O₂₁:Yb³⁺, Er³⁺ phosphors, *J. Alloys Compd.* 875 (2021), 159922.
- [46] C.S. McCamy, Correlated color temperature as an explicit function of chromaticity coordinates, *Color Res. Appl.* 17 (1992) 142–145.
- [47] Y.C. Li, Y.H. Chang, Y.F. Lin, Y.J. Lin, Y.S. Chang, High colour purity phosphors of LaAlGe₂O₇ doped with Tm³⁺ and Er³⁺, *Appl. Phys. Lett.* 89 (2006) 5–8.
- [48] S. Pradhan, H. Kaur, M. Simhadri, Photoluminescence and thermal sensing properties of Er³⁺ doped silicate based phosphors for multifunctional optoelectronic device applications, *Ceram. Int.* 47 (2021) 27694–27701.
- [49] M. Inokuti, F. Hirayama, Effect of energy transfer by the exchange mechanism on donor luminescence, *J. Chem. Phys.* 43 (1965) 1978–1989.